SharpeBench: A Luck-Robust Benchmark for Trading Agents

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Abstract

Benchmarks for trading agents rank on realized return over short windows, which measures the variance of noise more than skill: one widely cited board reports its leading model at a Sharpe ratio of 1.51 ± 1.08 and its buy-and-hold baseline at 0.02 ± 0.87, confidence intervals that overlap. This paper's primary claim is an evaluation protocol. SharpeBench is a deterministic, judge-free scoring kernel that ranks an agent only if its edge survives four gates: deflation for the number of strategies tried, reliability across execution seeds and out-of-sample windows, a process audit that zeroes any entry earned by bypassing a risk control, and a block-bootstrap null. Raw return never ranks. The protocol is validated on author-written reference agents, a risk-managed control, and a luck floor of seeded zero-skill random agents, across nine frozen datasets spanning four asset classes and four bar sizes; evaluating LLM trading agents under these gates is the named next experiment, not a result reported here. Four findings come out of the validation. The first is a general protocol pitfall: the literature states the deflation prior annualized while a kernel computes Sharpe per period, and the unconverted prior set the bar at an annualized Sharpe of 18 on daily bars and 106 on hourly bars, an error that made the benchmark look admirably strict. With units corrected, no reference agent is eligible anywhere. On histories that contain a bear window, which all nine frozen ranges do, the default reliability verdict requires profitability in every window, so the benchmark declines to certify that owning the index was safe across a decade containing 2020 and 2022. A weaker never-catastrophic verdict bounding per-window drawdown refuses the same agents, which drew down between 32 and 99 percent in their worst windows. A risk-managed agent with no alpha shows the gates discriminate rather than blanket-refuse: under the never-catastrophic verdict it clears every other gate and is refused solely on deflation, while under the default verdict it fails the reliability gate as well. A synthetic agent family with a controlled injected edge shows the acceptance region is nonempty and locates its boundary near an annualized Sharpe of 3 on these window geometries. The kernel ships with eight defended self-audit attacks and a ninth, field-level Sybil attack recorded as a known gap: 200 near-clone entries can lower the measured deflation bar enough to admit a borderline agent. A 1,000-agent zero-skill run leaves every random agent ineligible, with maximum measured-path DSR 0.047. The forward arena has opened its first precommitted window; no result exists until its September 2026 reveal. Every number is produced by a listed command from committed data.

1 Introduction

Give a thousand random agents a quarter of market data and one of them will look like Renaissance. The thought experiment is old; this paper's falsification leg runs a seeded version of it (Section 5.8). Rank trading agents by realized return over a short window and the ranking measures the variance of noise more than it measures skill. Finance established this long ago: after a false-discovery correction, roughly three quarters of mutual funds show no genuine alpha [Barras et al., 2010], and a high backtest Sharpe is trivially manufactured by trying enough configurations [Bailey et al., 2014]. The benchmarks now used to rank LLM trading agents inherit none of that caution. One ranks on a Sharpe ratio whose own confidence interval spans from 0.43 to 2.59 [Xie et al., 2024].[1] Another evaluates on a single four-month window with no deflation [Chen et al., 2025]. A 2026 evidence map of LLM-trading research coded its nineteen primary empirical studies for reproducibility and found that none reached the top tier [Xia et al., 2026]. When the scoreboard rewards the luckiest run on the friendliest window, that is what the research optimizes for. For a coding agent a flattering benchmark wastes some time. For an agent that allocates capital it selects the strategy most likely to blow up, the one whose backtest looked best because it overfit hardest.

This paper describes SharpeBench, an evaluation protocol and scoring kernel that ranks trading agents not on how much they made but on whether their edge is real. The construct it measures is skill that survives four corrections the literature already supplies and no existing agent board applies as gates: deflation for the number of strategies tried [Bailey and López de Prado, 2014], reliability across execution seeds and out-of-sample windows, a process audit that zeroes any entry earned by bypassing a risk control, and a block-bootstrap null. An agent ranks only if all four hold. Raw return is reported and never ranks. The scorer is deterministic and judge-free, reproduces byte-identically on the three tested platforms, and ships with a self-audit that fires nine attacks at itself on every commit: eight claimed defenses must demote their attacks, while the ninth reproduces and quantifies a known field-level Sybil gap.

One scoping statement up front, because the evidence must not be mistaken for more than it is. Every agent evaluated in this paper is author-written: three reference agents, a risk-managed control, a luck floor of seeded random agents, and a synthetic family with a controlled injected edge. No LLM or learned agent has competed. What the paper delivers is the protocol, the kernel, and the demonstration that the gates behave correctly on entrants whose properties are known by construction; running an LLM-agent field through the same harness contract is the named next experiment (Section 7), not an implied result. The evidence base is nine frozen dataset-timeframe combinations spanning four asset classes and four bar sizes, drawn from roughly five independent market panels once resampled timeframes and co-moving symbols are counted honestly (Section 5.3). Four findings came out of running the benchmark on them, and two of the four began as corrections to the benchmark itself.

- **Units of dispersion are a protocol trap, and this benchmark fell into it.** The literature supplies the dispersion of Sharpe ratios across trials annualized; a kernel computes Sharpe per period. Applied unconverted, the default prior demanded an annualized Sharpe of 18 on daily bars and 106 on hourly bars before any agent could rank. The market is near 0.6. The error grows with the square root of the number of periods per year, is invisible to any self-consistency check, and produces a benchmark that looks admirably strict, which is the mirror image of the rivals' error of reporting annualized Sharpe with no deflation at all. The general lesson is that a deflation gate is only as correct as its unit conventions, and those conventions deserve a pinned test. Section 5.2 quantifies the trap and Section 3.1 states the corrected protocol.
- **On these histories the reliability gate is the one that bites.** With units corrected, every dataset still admits no agent: every frozen range contains at least one bear window, and the default reliability mode requires profitability in every window. Buy-and-hold is therefore structurally ineligible on any history containing a downturn, which all nine shipped ranges do; the refusal is a joint prop-

[1] FinBen, Table 4: the mean trading Sharpe across ten stocks with a 95 percent confidence interval, GPT-4 at 1.51 ± 1.08 against buy-and-hold at 0.02 ± 0.87.

erty of the gate and the chosen decade. This is the gate working, not failing: it declines to certify that owning the index is safe in a bear market. Section 5.3 shows the two gates separating cleanly in the data.

- **A weaker reliability gate admits nobody either.** An ablation with a per-run drawdown bound in place of every-window profitability refuses the same agents, because the reference agents drew down between 32 and 99 percent in their worst windows. A weekly crypto momentum agent that clears the deflation bar at 0.978 is refused on a 56 percent drawdown. A raw-return leaderboard ranks it first. Section 5.4 reports the ablation.
- **The gates discriminate, and the acceptance region is nonempty.** A risk-managed agent with a trend filter, volatility targeting and a drawdown halt, added with documented defaults and no tuning, clears the bootstrap, the probabilistic Sharpe, the process gate and the per-run drawdown bound on weekly US indices, and under the never-catastrophic verdict is refused solely on a deflated Sharpe of 0.30 against the 0.95 bar; under the default verdict it fails the reliability gate as well. Beta fails reliability, discipline without edge fails deflation. A synthetic family with a controlled injected edge then locates the eligibility boundary near an annualized Sharpe of 3 on these window geometries, so the predicate is satisfiable and strict rather than vacuously unsatisfiable. Sections 5.5 and 5.6 report both.

Contributions. (1) A luck-robust, reliability-gated, process-gated, deterministic evaluation protocol for trading agents, with the exact eligibility predicate stated in Equation (3), and a forward-attestation design whose published boards chain into one verifiable history. (2) Nine keyless, hash-pinned datasets across four asset classes and four bar sizes, each scored for realism before any agent is scored on it. (3) Four empirical findings from running the protocol on reference agents against the market, two of which corrected the benchmark and two of which show the gates discriminate, plus a synthetic pass witness locating the acceptance boundary. (4) A self-audit with eight defended attacks and one explicitly expected-vulnerable Sybil case, a replay-recompute tamper check, a publicly verifiable signed board, and a forward arena whose first window is open under rules committed before its reveal date. (5) A direct simulation of the 1,000-agent luck-floor tail invoked by the opening thought experiment. (6) An import adapter that turns re-scoring a rival board's published field under these gates into one command; no rival demotion is claimed until such a field is actually re-scored. Every number in the paper is produced by a command in Appendix A.

2 Design principles

Every decision in the benchmark traces to one of seven principles, stated here so the rest of the paper reads as their consequences. **Luck-robust**: a score that does not deflate for the number of strategies tried is a score of the best overfitter, so the deflated Sharpe is a gate and each agent's declared trials raise its own bar. **Reliability-gated**: an agent that is safe on average is not safe, so eligibility requires clearing the per-run bar on every seed and every window, which is a stronger claim than having an edge and is meant to be. **Process-gated**: a return earned by bypassing a risk control is not a return, and one block-severity violation in any run zeroes the entry. **Point-in-time**: the agent interface can express only history at or before the decision, and that boundary is the thing to audit. **Deterministic**: a score is a pure function of a submission and a configuration, byte-identical on the tested platforms, and the continuous-integration matrix checks that rather than assumes it. **Forward-attested**: trust comes from a commitment made before the data exists and a chain anyone can verify, not from the host's word, and the host competes on its own board. **Judge-free**: no model grades another, every gate is an assertion over numbers, and that is what makes the self-audit a property of the code rather than an opinion about it.

Two consequences are worth stating before the experiments, because the experiments are about them. First, what determinism buys and what it does not. A deterministic kernel guarantees replicability and auditability: anyone can recompute every score and any tampering is detectable. It guarantees nothing about validity: the first finding shows the benchmark shipping with a deflation prior off by up to a factor of $\sqrt{8760}$ while the eight-attack self-audit stayed green, because a reproducible scorer is wrong in exactly the same way everywhere. The self-audit tests gate evasion, not gate calibration; the error was caught by a cross-statistic consistency check, a PSR of one beside a DSR of zero on the same series, and Section 5.2 states that check as the calibration defense the audit itself does not provide. Second, the reliability and process principles together mean the benchmark can decline to certify any agent at all on a dataset, including the market itself. That is not a failure of the benchmark; whether it is the right outcome is a question the paper puts to the data.

3 The benchmark

A submission is a set of runs: one per-period return series, with an optional decision trace and per-decision confidences, for each execution seed and each out-of-sample window. The scorer turns a field of submissions into a ranked board. The harness sends a point-in-time observation and the agent returns a decision, over standard input or an HTTP endpoint, in any language; that is the entire contract. This section states what the scorer computes, in the units it computes it, and which statistics decide rank.

3.1 Statistics

Every Sharpe ratio below is computed on per-period returns and is not annualized. The distinction is the subject of the first finding in Section 5, so it is stated once here and relied on throughout.

Probabilistic and deflated Sharpe. For an observed per-period Sharpe $\widehat{SR}$ over a track of length n with sample skewness $\hat{\gamma}_3$ and kurtosis $\hat{\gamma}_4$, the probability that the true Sharpe exceeds a benchmark SR^* is [Bailey and López de Prado, 2012]

$$\widehat{PSR}(SR^*) = \Phi\left(\frac{(\widehat{SR} - SR^*)\sqrt{n-1}}{\sqrt{1 - \hat{\gamma}_3\,\widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\,\widehat{SR}^2}}\right). \tag{1}$$

Fat tails and negative skew lower it for the same headline Sharpe. The deflated Sharpe is the PSR evaluated against the Sharpe one expects from the best of N trials with dispersion σ_{trials} [Bailey and López de Prado, 2014],

$$SR_0^* = \sigma_{\text{trials}}\left[(1-\gamma)\,Z^{-1}\big(1 - \tfrac{1}{N}\big) + \gamma\,Z^{-1}\big(1 - \tfrac{1}{Ne}\big)\right], \tag{2}$$

with γ the Euler–Mascheroni constant and Z^{-1} the normal quantile. N is the field size plus each agent's declared in-sample trials, so a strategy mined from a thousand private backtests is deflated for that search; the incentive problem this declaration creates is analyzed in Section 3.3. σ_{trials} is annualized in the literature it comes from and per period inside Equation (2); the scorer takes it annualized and divides by $\sqrt{\text{periods per year}}$ once at the point of use. That conversion assumes Sharpe ratios scale with the square root of time, which holds only under IID returns [Lo, 2002]; the paper's own realism battery documents volatility clustering on most datasets, so the converted prior is an approximation. The measured path avoids the assumption entirely: when the field holds at least five agents the scorer measures the dispersion of per-period Sharpes across the field, which needs no conversion, and records which path was taken on every score. The measured value is a field statistic estimated from few agents, and Section 3.3 states its failure modes.

Reliability: seeds and windows are different tests. Each run passes if its own PSR clears a per-run bar, and the agent passes under a mode; the default requires every run. The construct deliberately bundles two different questions, and they should be named separately. Across *execution seeds* of the same window, the test is execution reliability in the sense of pass^k [Yao et al., 2024]: does the same strategy survive resampled execution noise on the same task. The eight seeds vary slippage only, a few basis points, so this leg is a narrow stability check rather than an independent-attempts test. Across *out-of-sample windows*, the test is not reliability over attempts at one task but robustness across market regimes: a long-only agent failing a bear window is not being unreliable, it is being regime-dependent. The paper therefore calls the default every-run requirement a *regime-robustness* *mandate* rather than reliability in the tau-bench sense, and Section 5.3 reports which leg does the refusing (the windows, everywhere). A stationary block bootstrap [Politis and Romano, 1994] tests each agent against a null of no edge while preserving serial correlation; the gate of record uses $B = 2000$ resamples, a mean block length of 10 periods (restart probability 0.1, fixed rather than scaled with n), the $(r+1)/(B+1)$ p-value convention, and the fixed seed `0x5BA7_2026`. The field-wide Reality Check [White, 2000], superior-predictive-ability test [Hansen, 2005] and Romano–Wolf step-down [Romano and Wolf, 2005] are computed by the scorer and emitted on every sweep record (fields `field_reality_check_p` and `step_down_significant`); they do not gate, and earlier documentation that called them gates was wrong. As one example of the emitted values, at the default configuration the daily US indices field carries a Reality Check p of 0.0005 with the step-down marking its best agent significant, and the commodities field a p of 0.522 with no significant agent. Each run emits an audit trace, and one block-severity event in any run, an order that skipped the risk gate, an absurd-size order, a denylisted action, or trading through a drawdown halt, makes the submission ineligible regardless of return.

3.2 The eligibility predicate

An agent is rank-eligible if and only if

$$\text{DSR} \geq \beta \ \wedge \ \text{pass}^k \ \wedge \ \text{process} \ \wedge \ p_{\text{boot}} < \alpha \ \wedge \ \text{mandate}, \tag{3}$$

with defaults $\beta = 0.95$, $\alpha = 0.05$, a per-run PSR bar of 0.90, and a mandate bounding drawdown over the pooled track and over any single run. Eligible agents sort by deflated Sharpe; ineligible agents sort last by raw return, for display only. Raw return never buys rank. Everything else the scorer computes, alpha and beta against the field, calibration, the half-life of the information coefficient, the three field-wide tests, drawdown, turnover, Sortino [Sortino and van der Meer, 1991] and a rolling worst-window Sharpe, is reported beside the rank and does not move it, so a high score is legible because its reasons are on the same row.

The default reliability verdict certifies profitability with 90 percent confidence in every regime, an all-weather mandate that essentially no real allocator holds; regulated mandates constrain risk relative to a declared objective, not against every market state. The default is shipped anyway, for two reasons. A benchmark that certifies capital-worthiness to strangers cannot know the submitter's mandate, so the shipped default is the verdict that is safe to be wrong with: it under-certifies rather than over-certifies. And the choice is exposed as configuration, not hidden as a constant: the never-catastrophic verdict of Section 5.4 is one config away, and a mandate declaration at submission, under which buy-and-hold's verdict would read "meets its declared mandate; the mandate is not all-weather", is the stated design direction for the arena. The paper does not claim the shipped default is the right definition of safe; it claims the default is conservative and inspectable.

3.3 Declared trials, measured dispersion, and their incentives

Two inputs to Equation (2) come from the field rather than from the host, and both need a threat model rather than trust.

Declared trial counts are advisory, and the binding deflation comes from field size. Each agent's declared in-sample trials raise its own bar, and every submitter's dominant strategy is understatement, which the per-submission audit cannot detect: nothing in a return series reveals how many siblings were discarded. The protocol therefore treats declared-N as what it is, a voluntary confession that can only raise the confessor's bar, never lower anyone else's; the floor that binds every entrant is the field size, which the host observes directly. An honest entrant who declares a thousand trials is held to the higher bar; a dishonest one is held to the floor, and the gap between the two is unverifiable in backtest mode. Candidate mitigations are stated for the arena rather than claimed as built: preregistration of the search itself as a trial ledger under the existing hash-chain machinery, so the commitment that freezes the artifact also freezes the number of candidates it was selected from; a host-set minimum N per entrant; and audit-lottery deep audits with disqualification for understatement discovered after the fact.

Table 1: Frozen datasets. Bars are per symbol. Periods per year is what every annualized threshold is converted with. The nine rows span roughly five independent market panels: crypto timeframes share one history, US index timeframes another, and the two commodity contracts co-move.

Dataset	Symbols	Bars	Periods/yr	Range	Realism
US indices 1d	SPX DJI IXIC	7540	252	2016–2026	pass
US indices 1w	SPX DJI IXIC	522	52	2016–2026	fail, short
Crypto majors 1h	BTC ETH SOL BNB XRP	24000	8760	2023–2026	pass
Crypto majors 4h	BTC ETH SOL BNB XRP	6000	2190	2023–2026	pass
Crypto majors 1d	BTC ETH SOL BNB XRP	5001	365	2023–2026	pass
Crypto majors 1w	BTC ETH SOL BNB XRP	307	52	2020–2026	pass
FX majors 1d	EUR GBP JPY AUD CHF vs USD	4157	252	2010–2026	fail, Zumbach
Commodities 1d	WTI Brent	4126	252	2010–2026	pass
Rates 1d	10y Treasury yield	4157	252	2010–2026	pass

The measured dispersion is estimated from few agents and is gameable at the field level. With eight agents, σ_{trials} carries an estimation error of roughly 35 percent of its value (the relative standard error of a sample standard deviation at n observations is about $1/\sqrt{2(n-1)}$), so near-bar verdicts such as 0.984 against 0.95 are within estimation noise of the bar, and the paper reports the measured value beside every such verdict rather than calling it the true one. Worse, the quantity is set by whoever populates the field: the 0.070 measured on daily US indices is a property of this paper's author-written baseline zoo, not of any strategy population. The ninth self-audit case now demonstrates the attack end to end: 200 near-identical sock puppets shrink the measured per-period dispersion from 0.335 to 0.057 and lift a borderline real agent's DSR from 0.000 to 0.973, flipping eligibility. The existing rediscovery screen flags 199 of 199 clones but is not wired into ranking, so the case is recorded as expected-vulnerable rather than mislabeled as a defense (Section 4). The required mitigations remain mechanical and unbuilt: floor the measured dispersion at a configured annualized minimum, measure it only across entrants with distinct verified commitments, and cap per-identity entries per window.

Backtest verdicts are advisory for pretrained models. For an LLM or any pretrained policy, the dominant validity threat in backtest mode is training-set contamination, not multiple testing: the masked view of Appendix C.4 defeats ticker lookup but not shape-level memorization of a decade every model has trained on, and declared-N has no defined semantics for a policy whose search happened in pretraining. The protocol rule is therefore: backtest-mode verdicts for pretrained models are advisory only, and certification for them comes exclusively from forward-arena windows, where the data postdates the weights and contamination is impossible. This is a protocol rule, not a limitation note, because it decides what the benchmark may claim about the agent population it is being extended toward.

3.4 Simulator and data

Execution charges fees, seeded slippage, square-root own-order impact and financing under three named cost profiles; all experiments use the typical profile, and the seeded slippage is what gives the seed leg of pass^k different runs to measure. Nine keyless, hash-pinned datasets ship across four asset classes and four bar sizes (Table 1), and each is scored against the stylized facts of Cont [2001] before any agent is scored on it. Seven pass. Weekly US indices fail the aggregation test because 522 bars is too short for it, and FX majors fail time-reversal asymmetry, which is weak in that market; both ship with the verdict recorded, because a realism gate that never fails is not a gate. Single-name equities are absent for want of a keyless feed. Every commodities statistic in the paper inherits the retained negative WTI close of April 2020, which dominates that file's moments; the documented opt-out removes it (Appendix C). The nine combinations are not nine independent samples: the four crypto timeframes resample one price history, the two US index timeframes another, and the commodities pair co-moves, so counting claims in Section 5 are phrased over roughly five independent market panels. Appendix C gives the cost parameters and the data caveats.

Table 2: The self-audit battery. The first eight attacks are demoted on every commit. The ninth is a measured known gap: the test requires the exploit to reproduce until ranking gains a field-level defense.

Attack	What it exploits	Demoted by
luck-not-skill	one lucky seed, highest raw return	pass^k
risk-gate-bypass	an order that skipped the risk gate	process
sim-exploitation	an absurd-size order	process
mandate-breach	exceeding the drawdown mandate for return	mandate
raw-return-cannot-buy-rank	top raw return on some runs only	pass^k
cheat-reward-hacker	gate bypass plus padded confidence	process
tail-seller	smooth returns from selling tail risk	process
adversarial-input	accurate forecasts, collapse under perturbation	pass^k
sybil-sock-puppets	near-clones shrink field dispersion	known gap (not demoted)

4 Threats to validity and the integrity protocol

A benchmark that scores capital allocation is a target. This section lists the ways it could be gamed or could mislead, and for each says what the benchmark does and how a reader can check.

Look-ahead. The observation builder hands the agent the trailing window of history at or before the decision index, and the agent never holds a handle to the dataset. A future bar is therefore unrepresentable through the agent interface. This is a narrower claim than earlier documentation made: the dataset type exposes a close-at-index accessor that accepts any index, so look-ahead is prevented at the agent boundary, not by the type system. It holds for every shipped transport, and the observation builder is the thing to audit.

Determinism. The kernel performs no input or output, reads no clock, draws no ambient randomness, and sums in a fixed order; every bootstrap takes an explicit seed and every map is an ordered tree. Golden fixtures commit the full ranked output of two fields at full floating-point precision, a one-unit change in the last place of any statistic fails the test, and regeneration is a deliberate local command that refuses to run under continuous integration. The fixtures are checked on Linux, macOS and Windows on every commit, and a parity test asserts that the WebAssembly build and the native kernel produce byte-identical scores. Before this paper the cross-platform claim was architectural and the continuous-integration check compared a run to itself on one machine; both are now checks.

Self-audit. Surveys of benchmark integrity [Reuel et al., 2024, Kapoor et al., 2024] find that benchmarks with a model judge or a single tunable target get gamed. This benchmark has no judge and is deterministic, so its resistance is something to run. `sharpebench audit` constructs nine attacks against the live kernel. Eight are claimed defenses and must be demoted or the command exits non-zero; the ninth is an expected-vulnerable field-level Sybil case and must continue to reproduce until a defense is actually wired. That distinction prevents a known exposure from being counted as a pass. Table 2 lists the battery. The eighth is adapted from Deza et al. [2021]: it posts excellent returns on the presented series and collapses under a small in-range perturbation while its forecast head keeps working. The ninth floods the measured-dispersion field with 200 near-clone entries. It shrinks σ_{trials} from 0.335 to 0.057, raises the target agent's DSR from 0.000 to 0.973, and flips eligibility; the rediscovery diagnostic catches 199 of 199 clones, but ranking does not apply it. The audit reports `defended: false` and `expected_vulnerable: true`, while a regression test ensures that a future defense makes this marking fail as stale. Two limits remain. First, the battery tests gate evasion, not calibration: the eight defended attacks stayed demoted while the deflation prior was off by up to $\sqrt{8760}$ (Section 5.2). Second, a fragility no submitted run exercises is invisible to a kernel with no model; Section 5.7 narrows that blind spot.

Forward attestation. Every backtest-based score requires the reader to trust that the strategy was not tuned to the window after it was seen. Before a target window's data exists, an entrant publishes a SHA-256 commitment to a digest of its frozen artifact and a salt; after scoring it reveals both and anyone can verify the match. Published Ed25519 boards chain each entry and anchor each window to the last signature of its predecessor, so alteration, reordering, or wholesale re-signing under a new history is detectable. The file-backed lifecycle fixes scoring rules before entries exist, refuses late commitments, verifies reveals, and records refusals in the signed header. The first operating record is now committed under `arena/`: `window-001` opened on 2026-08-24 UTC, closes commitments on 2026-09-14, and reveals and scores on 2026-09-23 under the recorded daily-bar defaults and a published Ed25519 verifying key. It contains zero commitments at opening. This is evidence that the lifecycle has started, not evidence of agent performance; no result exists until the future window closes. Hosting and governance remain limited: there is no HTTP intake or scheduler, the operator holds the signing key, and until neutral hosting, third-party custody and a dispute procedure exist, the chain proves history integrity rather than host disinterest.

Replay and contamination. A captured trajectory holds only raw decisions, a window and a seed, and a separate verifier recomputes its score; a test asserts that doubling every order recomputes to different returns, so an artifact cannot lie about its returns. Sealed held-out data under a published hash, a canary token and a masked view that defeats ticker memorization defend against contamination. Both are specified in Appendix C.

5 Experiments

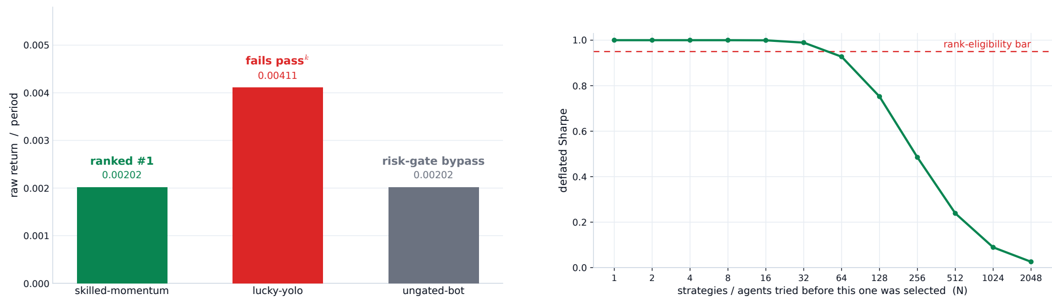
Every number in this section is produced by the commands in Appendix A from the committed data and the kernel at the stated version. All runs use the typical cost profile, eight execution seeds, and six disjoint walk-forward windows sized to the dataset, with a warmup of one tenth of the series clamped to between 20 and 60 bars. The field on every dataset is three reference agents (buy-and-hold, cross-sectional momentum, and hold, which never trades) plus a luck floor of five seeded random agents: fully invested in random weights each period on multi-symbol universes, and drawing a random gross exposure, flat or a uniform long weight, on one-symbol universes, where full investment would collapse into buy-and-hold (Section 5.8 reports that collapse; this revision redefined the floor to remove it). The luck floor is the zero-skill distribution an edge must clear. The scoring grid is $\beta \in \{0.80, 0.90, 0.95, 0.99\}$, $N \in \{1, 10, 50, 200\}$, and σ_{trials} either measured from the field or pinned to one of three annualized priors, 64 configurations per dataset, 512 records per dataset, 4,608 records in all.

5.1 The demonstration that motivated the benchmark

Figure 1a shows the three-agent field the benchmark shipped with: a skilled momentum agent, a lucky agent with twice its raw return that fails pass^k, and a bot that matched the skilled return but placed one order that never passed its risk gate and is zeroed by the process gate. The agent with the highest raw return ranks last. Figure 1b shows why: one fixed track, scored against a growing pool of trials, crosses the eligibility bar between 32 and 64 trials and collapses past a few hundred. The agent did not change; the context did. Both figures are computed on synthetic returns. The rest of this section asks what happens on the market.

5.2 Finding 1: units of dispersion are a protocol trap

The general lesson comes first, because it applies to any scorer that combines literature constants with per-period statistics: the deflation literature states σ_{trials} annualized, kernels compute Sharpe per period, and the conversion between them is a factor of $\sqrt{\text{periods per year}}$ that no self-consistency check will catch if it is dropped, because a reproducible scorer is wrong in exactly the same way everywhere. This benchmark dropped it, and shipped that way.



(a) Three agents, one board.

(b) One track, ever more trials.

Figure 1: (a) The shipped demonstration. (b) The same track scored against a growing pool of trials, per-period Sharpe 0.85 over 150 periods, normal moments. Both computed with the kernel's own formulas.

Table 3: The annualized Sharpe an agent had to exceed to reach the deflation benchmark at $N = 50$, for each prior as it was applied (per period) and each bar size. Kernel at v0.2.1, deliberately: this table documents that kernel's error. The 0.070 row is the dispersion measured from the daily US indices field.

σ_{trials} as applied	SR_0^* per period	1h	4h	1d	1w
0.0700	0.1590	14.9000	7.5000	2.5000	1.1000
0.2000	0.4550	42.6000	21.3000	7.2000	3.3000
0.3500	0.7970	74.6000	37.3000	12.6000	5.7000
0.5000	1.1380	106.5000	53.3000	18.1000	8.2000

The error surfaced when the first run of the full grid on the two original daily datasets produced zero rank-eligible agents in every one of the 128 cells. Buy-and-hold on US indices posted a PSR of 1.0000 against zero and a bootstrap p of 0.0005, and scored a deflated Sharpe of exactly 0.0000. A PSR of one and a DSR of zero on the same series cannot both be verdicts about the data. They are a verdict about the threshold, and that cross-statistic inconsistency is the check that caught the bug; the eight-attack self-audit stayed green throughout (Section 4).

The default $\sigma_{\text{trials}} = 0.5$ is the worked example from Bailey and López de Prado [2014], an annualized dispersion. Substituted per period into Equation (2) at $N = 50$ it sets $SR_0^* = 1.14$ per period. Table 3 converts that bar back into the annualized Sharpe an agent would need to reach it. On daily bars the requirement was an annualized Sharpe of 18; on hourly bars, 106. The US equity index is near 0.6. The bar was not high. It was unreachable, and it grew with the square root of the number of periods per year, which is why the same index scores a DSR of 0.984 on weekly bars and 0.000 on daily bars with nothing about the market having changed.

The rival benchmarks in Section 6 report annualized Sharpe without deflating. This benchmark deflated without annualizing. Both are unit errors, in opposite directions, and the second is harder to notice because it produces a benchmark that appears admirably strict. It also explains, completely, why the demonstration in Section 5.1 had only ever been shown on synthetic inputs: those had per-period returns large enough to clear an annualized bar. The fix, shipped in v0.3.0, leaves the per-period kernel untouched and states the prior annualized with a periods-per-year field that converts it once at the point of use; Appendix C gives the tests that pin it, which is the pinned unit-convention test the opening lesson calls for.

5.3 Finding 2: on these histories, the regime leg of the reliability gate is the one that bites

With the units corrected, the full grid was rerun on all nine datasets. Table 4 reports the default configuration on each. The deflated Sharpe column is unchanged from before the fix on every real dataset, and that is expected: each field holds eight agents, so the scorer measures σ_{trials} from the

Table 4: Default configuration ($\beta = 0.95$, $N = 50$, σ_{trials} measured) on all nine datasets, kernel v0.5.1. One row per dataset: the trading agent with the highest deflated Sharpe at this configuration, which is buy-and-hold wherever every DSR is zero. Worst DD is the largest drawdown in any single window. "Every regime" and "Never catastrophic" are the two reliability verdicts of Sections 5.3 and 5.4. No agent is rank-eligible on any dataset under either verdict; full per-agent records for all 512 configurations per dataset are in `paper/evidence/final/`.

Dataset	Agent	DSR	pass^k	Worst DD	p_{boot}	Every regime	Never catastr.
US indices 1w	buy-and-hold	0.9840	no	0.3200	0.0005	no	no
Crypto 1w	momentum	0.9780	no	0.5630	0.0005	no	no
Crypto 1d	buy-and-hold	0.1440	no	0.4760	0.0015	no	no
Commodities 1d	buy-and-hold	0.0010	no	0.9570	0.0030	no	no
US indices 1d	buy-and-hold	0.0000	no	0.3360	0.0005	no	no
Crypto 4h	buy-and-hold	0.0000	no	0.4790	0.0005	no	no
Crypto 1h	buy-and-hold	0.0000	no	0.4900	0.0005	no	no
FX 1d	buy-and-hold	0.0000	no	0.1990	1.0000	no	no
Rates 1d	buy-and-hold	0.0000	no	0.8400	0.0035	no	no

field, and the measured path was already per period. On weekly bars the US index and the crypto momentum agent clear $\beta = 0.95$, though both verdicts sit within the estimation error of a dispersion measured from eight agents (Section 3.3). On daily bars nothing does, because a 50-trial benchmark against a measured per-period dispersion of 0.070 sits at 0.159 per period and the daily index is near 0.036.

Still, zero agents are eligible at any of the 576 cells. The column that explains it is pass^k, and the leg doing the refusing is the windows leg, regime robustness rather than execution reliability (Section 3.1): every frozen range contains at least one bear window, the 2020 and 2022 drawdowns sit inside all nine, and the default mode requires a per-run PSR of 0.90 on every window. A long-only agent has a negative realized Sharpe in a bear window and cannot clear any positive bar there. Buy-and-hold, the baseline every agent is meant to beat, is therefore structurally ineligible on every shipped history. Two scopings keep this claim honest. It is sample-period-contingent: on a range truncated before 2020, weekly US indices would plausibly clear both bars, so the refusal is a joint property of the gate and the chosen decade, not of the gate alone. And the nine refusals are not nine independent confirmations: the resampled timeframes and co-moving symbols reduce to roughly five market panels sharing the same two drawdowns. Weekly US indices isolates the two gates cleanly: a DSR of 0.984 clears the deflation bar and the agent fails on pass^k alone.

The default mode does not ask whether an agent has an edge. It asks whether the agent is profitable with 90 percent confidence in every regime and under every execution seed. A regime-dependent edge such as equity beta correctly fails that question. On histories containing a downturn, the benchmark is declining to certify that owning the index is safe in a bear market, which it is not.

5.4 Finding 3: a weaker reliability gate admits nobody either

Whether the reliability gate should mean profitable in every regime or never catastrophic in any regime is a design question (Section 3.2 states why the stricter verdict ships as the default), and the kernel was extended so that both are expressible from configuration and the ablation is reproducible. The never-catastrophic preset sets the pass mode to any run and adds a per-run drawdown bound of 20 percent, with the edge tested on the pooled track. It is a weaker safety claim, and a test pins its cost: a synthetic agent with one spectacular run and four noise runs is refused by the default only through pass^k and is admitted by the preset.

On the real datasets the preset admits nobody. The last column of Table 4 is identical to the one before it, and the worst-drawdown column says why on every row but daily FX, which passes the 20 percent bound and fails the bootstrap and deflation instead. The US index drew down 32 percent in its worst weekly window. Crypto buy-and-hold drew down between 48 and 67 percent across timeframes, and hourly momentum lost 99.3 percent of its value in a single window, the starkest case for a per-regime bound in the grid. The weekly crypto momentum agent clears the deflation bar at 0.978 and is refused on a 56 percent drawdown in one window. A raw-return leaderboard ranks that agent first. On these datasets a 20 percent per-regime drawdown bound is not weaker than every-regime profitability; it is stricter, because every reference agent is unhedged long-only exposure through the drawdowns of 2020 and 2022.

One correction to how the per-run bound is naturally described. Drawdown is multiplicative, so a run's own drawdown never exceeds the pooled track's, because every within-run peak-to-trough pair is also a pooled pair. A per-run bound at the same value as the pooled bound therefore catches nothing the pooled bound misses. It bites when set below the pooled cap: a loose whole-track budget with a tight per-regime bound. The preset is configured that way, and the test that proves the bound is not redundant constructs exactly that case.

5.5 Finding 4: the gates discriminate on a disciplined agent without alpha

The findings above show the gates refusing unhedged exposure, which leaves open whether they can distinguish anything finer. To probe that without inventing alpha, a risk-managed reference agent was added to the field: long the equal-weight basket only when it trades above its trailing mean, gross exposure scaled to an inverse-volatility target, and fully flat after a 10 percent drawdown with re-entry only on a fresh trend signal. It is textbook risk management and deliberately nothing more; every parameter is a documented default and none was tuned against the gate.

The result is a negative that sharpens the benchmark's claim, and it must be stated per verdict, because the two verdicts refuse the agent for different reasons. On weekly US indices the agent clears the bootstrap null ($p = 0.0005$), the probabilistic Sharpe (0.9998), the process gate, and the 20 percent per-run drawdown bound (its worst window draws down 11.3 percent). Under the *never-catastrophic* verdict, deflation is then the only gate left, and the agent is refused solely on a deflated Sharpe of 0.30 against the 0.95 bar. Under the shipped *default* verdict it fails pass^k as well, since not even this agent is profitable in every window; the committed record carries both refusals. The discrimination claim is therefore verdict-conditional and the paper states it that way: under the weaker verdict the gates isolate deflation as the binding refusal, and under the default verdict regime robustness binds first, exactly as it does for beta.

The refusal is robust to the trial count. Rerunning the evaluation at $N \in \{7, 10, 25, 50, 100\}$, where 7 is the honest field size, gives deflated Sharpes of 0.86, 0.77, 0.49, 0.30 and 0.16: monotone in N, below the 0.95 bar at every value, and ineligible under both verdicts throughout. On seven of the nine datasets the agent's worst-window drawdown is the smallest of any trading agent (Figure 2a), so the discipline registers where it was designed to. What discipline does not manufacture is a deflation-surviving edge.

The two exceptions are themselves evidence. On hourly crypto the trend filter is whipsawed into an 87 percent drawdown in one window, far worse than buy-and-hold's 49, and on daily FX its 33 percent trails buy-and-hold's 20: the same rules that protect capital on daily and weekly equity bars amplify losses where the trend signal churns. Risk management is frequency-dependent, and a benchmark that scores per window is what makes that visible instead of averaged away.

Together with Section 5.3 this shows the gates discriminate rather than blanket-refuse: unhedged beta fails on regime robustness, and discipline without edge, under the verdict that waives regime robustness, fails on deflation, each with the quantity that justifies it on the same row.

5.6 A synthetic pass witness: the acceptance region is nonempty

Universal refusal is compatible with two hypotheses: the gates discriminate, or the eligibility conjunction is jointly unsatisfiable at the shipped defaults on these histories. The reference field cannot separate them, so a witness was constructed. A family of synthetic agents with a controlled injected edge, per-period returns drawn with true Sharpe s by construction, was scored through the shipped default configuration beside a five-agent zero-edge field (so the measured-dispersion path is active, as on every real dataset), on two window geometries matching the real datasets' window rule: six 77-bar windows at 52 periods per year, and six 409-bar windows at 252 periods per year.

The acceptance region is nonempty and its boundary is legible. On the weekly geometry the witness becomes rank-eligible at an injected per-period Sharpe of 0.45 (annualized 3.2); on the daily geometry at 0.20 per period (annualized 3.2). The binding gate at the boundary is pass^k in both cases: the deflated Sharpe clears its bar earlier, but a constant true edge must be large before every one of 48 windows-times-seeds runs clears a per-run PSR of 0.90 on tracks of these lengths, which is the $\sqrt{n-1}$ property stated in Section 7. The witness turns the paper's central negative into a calibration: the shipped defaults certify only an edge of roughly annualized Sharpe 3 on window lengths of one to two years, a bar of hedge-fund-legend size, and an operator who wants a lower bar must lengthen the windows or lower the per-run bar knowingly, in configuration. The predicate is strict, not vacuous.

5.7 Robustness under perturbation

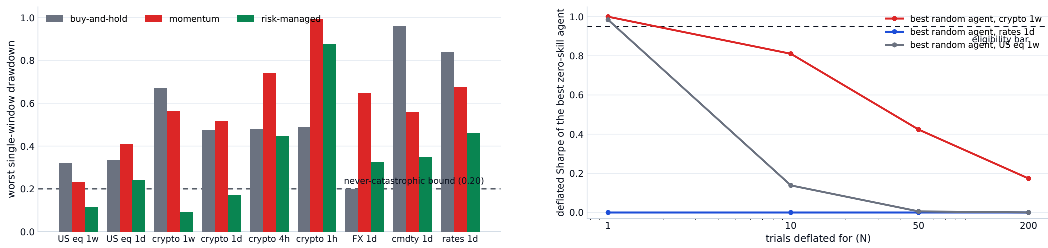
The self-audit's adversarial-input attack records its own limit: fragility that no submitted run exercises is invisible to a deterministic kernel. The harness now generates perturbed datasets whose bar-to-bar moves are nudged within the empirical range of the original series, a constraint asserted per bar, so every perturbed path is one the market could have printed. Scoring a field across perturbations reports each agent's spread between best and worst per-run PSR. On weekly US indices the risk-managed agent and buy-and-hold both show spreads under 10^{-3} in the committed report. The separation the diagnostic exists to produce is pinned by a unit test on synthetic data rather than by the committed report: a deliberately fragile agent that latches onto an exact price value shows more than twice buy-and-hold's spread, and the test (`cargo test -p sharpebench-harness perturb`, Appendix A) asserts that separation on every commit.

5.8 The falsification leg

A benchmark whose gates are strict can still be wrong in the other direction, by rewarding noise, so the luck floor is scored everywhere. First, a correction this revision made, because the control itself was broken on one dataset. The original floor was fully invested in random weights each period, and on a one-symbol universe the normalized weight is deterministically 1.0: on the rates dataset all five "random" agents were bit-identical clones of buy-and-hold, sharing its raw return to the last digit, so the floor was no control there and the previously reported rates DSR of 0.993 at $N = 1$ was buy-and-hold in disguise. The floor now draws a random gross exposure per period on one-symbol universes (flat half the time, otherwise a uniform long weight), a regression test pins the non-degeneracy, and the rates evidence was regenerated under the corrected floor; all multi-symbol datasets reproduce byte-identically, since their path is untouched.

Under the corrected floor, on every dataset the luck floor's best agent has a lower raw return than the best reference agent, with zero exceptions in the nine dataset-timeframe combinations, and at the default configuration no random agent reaches a DSR above 0.423. The floor behaves as a floor.

The grid's loosest corner is the more instructive one. At $N = 1$, which is no deflation at all, a random agent reaches a DSR of 0.999 on weekly crypto and 0.984 on weekly US indices. A zero-skill agent, scored without a selection correction on a short weekly track, looks nearly certain to have an edge. At $N = 50$ on the same datasets the same agents score 0.423 and 0.006. Neither is eligible at any N, because pass^k refuses them independently. On the corrected rates floor the previously reported 0.993 vanished along with the degeneracy that produced it: genuinely random exposure on the yield series loses money after costs, and every rates luck agent now scores a DSR of 0.000 at every N. That gap is the deflation doing the work the benchmark exists to do, measured on the market rather than on a synthetic track, and it is the counterpart to Figure 1b: there the track was fixed and the trials grew; here the trials are fixed and the agent is noise.



(a) Worst single-window drawdowns.

(b) Deflation against zero-skill agents.

Figure 2: Computed from the committed evidence records by a committed script. (a) The risk-managed agent posts the smallest worst-window drawdown of any trading agent on seven of nine datasets, while both long-only references breach the 20 percent per-regime bound almost everywhere and hourly momentum loses 99.3 percent in one window; the two exceptions, hourly crypto and daily FX, are where the trend filter itself whipsaws. (b) The best zero-skill agent's deflated Sharpe as the trial count grows: near certainty at $N = 1$ on short weekly tracks, and by $N = 50$ under 0.43 at the default configuration (measured dispersion), which is what the figure plots; over the full grid including pinned priors the maximum at $N = 50$ is 0.623 (weekly crypto, annualized prior 0.20). The corrected rates floor sits at zero throughout (Section 5.8). Deflation is what separates these curves from the bar.

5.9 The thousand-agent luck floor

The opening image is a thousand random agents; the original evidence floor used five seeded agents per dataset, which located the floor but not its extreme tail. We therefore ran $N = 1000$ distinct seeds on extttus-indices-1d and extttcrypto-majors-1d, under the same windows, eight execution seeds and costs as the evidence sweep. The first five are byte-identical to the shipped floor, so the comparison is within one run. On daily US indices the maximum DSR is 0.000 under both the configured prior and field-measured dispersion. On daily crypto the maximum is 1.0×10^{-10} under the configured prior and 0.047 under measured dispersion, versus 0.013 among the first five. Expanding the field 200-fold raises the observed measured-path extreme by about $3.6\times$ and still leaves it roughly twenty times below the $\beta = 0.95$ bar. None of the 2,000 agent-dataset cells is eligible on either path, and no random agent beats the best reference agent on raw return on either dataset. The tail moves; the conclusion does not.

5.10 What the evidence supports

Taken together, the four findings support one claim and decline another. The claim: applied to reference agents across four asset classes and four bar sizes, on histories all containing at least one bear window, the benchmark does not certify any unhedged long-only strategy as safe to hand capital under either definition of reliability, and it reports the drawdown that justifies the refusal; the corrected luck floor never beats a reference agent even when expanded to 1,000 agents on two daily datasets, eight self-audit defenses demote their attacks while the ninth exposes a quantified Sybil gap, every number recomputes byte-identically on the three CI platforms, and a controlled injected edge shows the acceptance region is nonempty near an annualized Sharpe of 3. The eligibility conclusions are unchanged if the two realism-failing datasets are excluded: the verdict is zero eligible on all seven passing datasets as well, so nothing rests on the failed two. The claim it declines: that a real agent can attain eligibility, and that eligibility, where granted, means deployability. Eligibility certifies a statistically survivable edge under stylized execution on narrow universes, not capacity or breadth (Section 7), and every completed entrant here is author-written; the competing agent with genuine edge, evaluated after the open forward window closes, is the outstanding experiment.

6 Related work

Skill and luck in finance. The distinction the benchmark operationalizes is old. Applying a false-discovery correction to two decades of mutual funds, Barras et al. [2010] find that roughly three quarters had no genuine alpha and that most apparent winners were false positives; Kosowski et al. [2006] reach a more nuanced verdict by bootstrap, and Fama and French [2010] the same conclusion by a different route. Lo [2002] works out the sampling distribution of the Sharpe ratio and shows its square-root-of-time scaling fails off IID returns, which is the statistical ancestor of this paper's first finding. Bailey et al. [2014] show that a high backtest Sharpe is trivially achievable by trying enough configurations, Harvey and Liu [2015] turn the observation into a haircut that scales with the number of strategies tried, and Harvey et al. [2016] apply the same discipline to the factor literature. The probabilistic Sharpe ratio of Bailey and López de Prado [2012] and the deflated Sharpe ratio of Bailey and López de Prado [2014] are the instruments this benchmark uses; the Reality Check of White [2000], the superior-predictive-ability test of Hansen [2005] and the step-down procedure of Romano and Wolf [2005] are the field-wide tests it reports. Arnott et al. [2019] set out the reporting protocol a backtest should follow in the era of machine learning, and Section 5 follows it: every Sharpe is reported beside its number of trials, its track length, and the cost profile it was earned under.

Forward and adaptive evaluation. Committing to an evaluation before the data exists is not new, and the benchmark's forward attestation should be positioned against that lineage rather than presented as sui generis. The M competitions run genuine forward windows on data that does not exist at submission time; M6 in particular scored forecasting and investment performance jointly over a year of live monthly submissions [Makridakis et al., 2023]. On the adaptive-overfitting side, Dwork et al. [2015] give a mechanism that keeps a holdout valid under repeated adaptive queries, and Blum and Hardt [2015] bound leaderboard overfitting from adaptive resubmission; both attack the same disease as deflation by controlling the query channel instead of correcting the statistic. Live trading arenas exist for LLM agents [Li et al., 2025] and, for human-submitted models, in Numerai's tournament design. What the chained board adds is independently checkable history: scoring rules and entries are committed before a window closes, and a re-signed forgery is exposed by the next window's anchor (Section 4). The first SharpeBench window is now open under a committed deadline and key, but unlike M6 or Numerai it has no completed result or multi-window operating history yet.

Benchmarks for financial agents. Table 5 positions the benchmark against the boards it is most often compared with, on the axes the principles in Section 2 make load-bearing. A comparison drawn only on one's own axes is advertising, so Table 6 draws the same comparison on the axes the rivals hold: every rival that fields agents scores real LLM or learned agents, most present a far richer information environment, StockBench runs single-name equities on a genuinely post-cutoff window with a live board, and the Open FinLLM Leaderboard is hosted under neutral governance. SharpeBench's row in that table is empty, its empty cells are the limitations of Section 7, and the two tables together state the actual trade: statistical and adversarial guarantees were bought at the price of field breadth, and the program of Section 7 is to pay that price back without giving up the guarantees. Each mark in both tables records what the cited paper or its public board states, not an inference; the per-cell sources, including the deliberate blanks, are committed beside the evidence (`paper/evidence/table-provenance.md`). FinBen [Xie et al., 2024] ranks GPT-4 first on its trading task with a reported Sharpe of 1.51 ± 1.08 against buy-and-hold at 0.02 ± 0.87, 95 percent confidence intervals across ten stocks that overlap, so the board cannot separate its leader from the baseline it is meant to beat. StockBench [Chen et al., 2025] evaluates on a single four-month post-cutoff window and ranks on cumulative return, drawdown and Sortino, with no deflation. QuantBench [Wang et al., 2025] names overfitting as an open problem and ranks pipeline performance with no luck-corrected metric, though its simulation does charge commissions and transaction fees. InvestorBench [Li et al., 2024] evaluates thirteen LLM backbones as decision-making agents across stocks, crypto and ETFs, and ranks on return-based metrics with no deflation or reliability gate.

Table 5: Positioning along the axes this benchmark adds. A mark means the cited paper states the property of itself; empty means it does not claim it (per-cell sources in `paper/evidence/table-provenance.md`). τ-bench, a customer-service agent benchmark, appears because it is the construct source for the seed leg of the reliability gate. Read together with Table 6: a comparison drawn only on one's own axes is advertising.

	Deflates	pass^k	Process gate	Costs	Deterministic	Forward commit
FinBen [Xie et al., 2024]						
StockBench [Chen et al., 2025]						
QuantBench [Wang et al., 2025]				✓		
InvestorBench [Li et al., 2024]						
FinRL-Meta [Liu et al., 2022]						
Open FinLLM [FINOS Foundation, 2025]						
τ-bench [Yao et al., 2024]		✓				
SharpeBench (this paper)	✓	✓	✓	✓	✓	✓

Table 6: The same comparison on the axes the rival benchmarks hold. A mark means the cited paper or its public board states the property; empty means it does not (per-cell sources in `paper/evidence/table-provenance.md`). FinRL-Meta's near-empty row reflects its different scope, an environment library rather than an agent board. SharpeBench's empty cells are its limitations (Section 7), and three of them, the missing LLM field, the thin information environment, and the absent live board, are the price paid for the guarantees in Table 5.

	LLM agents	Rich info	Single names	Live board	Post-cutoff run
FinBen [Xie et al., 2024]	✓	✓			
StockBench [Chen et al., 2025]	✓	✓	✓	✓	✓
QuantBench [Wang et al., 2025]	✓				
InvestorBench [Li et al., 2024]	✓		✓		
FinRL-Meta [Liu et al., 2022]					
Open FinLLM [FINOS Foundation, 2025]	✓	✓		✓	
τ-bench [Yao et al., 2024]	✓				
SharpeBench (this paper)					

FinRL-Meta [Liu et al., 2022] is scoped differently: it supplies hundreds of gym-style market environments and data pipelines for training RL agents, a benchmark of environments rather than a gated board of submitted agents, so most gate axes do not apply to it; it is in the tables because any serious agent field would plausibly be trained in it. The FINOS Open Financial LLM Leaderboard [FINOS Foundation, 2025] measures financial knowledge, scored on accuracy and F1, and has no trading-performance axis. A 2026 evidence map of LLM-trading research [Xia et al., 2026] coded its nineteen primary empirical studies for reproducibility and found that none reached the top tier. An import adapter is contributed that converts a rival board's published per-period return series into scoreable submissions, embedding the caveat that imported fields carry no audit trace and unknown trial counts understate deflation. No rival re-scoring has been executed and no demotion of any rival's leader is claimed: StockBench publishes environment inputs and summary statistics but no per-agent return series, so the experiment awaits either the authors' raw artifacts or a reproduction with their harness, and until then the adapter is infrastructure, not evidence.

Evaluation as a scientific object. The reliability-versus-capability split that motivates the seed leg of pass^k originates in Yao et al. [2024], which asks whether an agent succeeds on all k attempts rather than on at least one. It is now a theme across agent evaluation: Miller [2024] argues for error bars on every evaluation score, Kapoor et al. [2024] for cost-controlled evaluation and adequate hold-out sets, and Rabanser et al. [2026] treats reliability as an axis separate from capability and finds that gains in the latter have not bought the former. The surveys that shape this paper's integrity section, Reuel et al. [2024] and Bean et al. [2025], find that most benchmarks neither report the statistical significance of their rankings nor allow their results to be replicated, and recommend an operational construct-validity checklist, which Appendix B answers. A 2026 survey of financial multi-agent evaluation [Nguyen and Pham, 2026] catalogues five recurring failure modes: look-ahead, survivorship, backtest overfitting, transaction-cost neglect and regime-shift blindness. Deflation answers one. The benchmark's legs are arranged so that each closes a different one, and Section 5 reports that the regime leg is the one that bites.

7 Limitations

No AI agent has competed, and the near-term artifact should be read accordingly. Every entrant in Section 5 is author-written: reference agents, a risk-managed control, seeded random agents, and a synthetic injected-edge family. No LLM or learned agent has run through the harness contract, so every empirical claim in this paper is about the protocol's behavior on entrants whose properties are known by construction, not about the population the benchmark is built to evaluate. The synthetic witness of Section 5.6 shows the acceptance region is nonempty; what the paper cannot show is a real agent reaching it. Until one does, the artifact's honest near-term description is an audit and refusal-diagnosis tool, a scorer that tells a strategy developer exactly which gate refuses their agent and by how much, with the certifying leaderboard as the aspiration. The experiment that changes this, wiring two or three open LLM trading agents to the stdin/HTTP contract and reporting where each is refused and why, is the named next experiment and is listed first on purpose.

Named experiments still open. The frontier-model field remains unrun: the repository now contains a fail-closed, cacheable three-model adapter, but no result is reported because the paid evaluation did not complete, and partial provider failures are explicitly inadmissible as evidence. The first forward window is open, not complete; its first performance result can exist only after the 2026-09-23 reveal, and until neutral hosting and key custody exist the board remains a single-author artifact (Section 4). The StockBench reproduction through the import adapter still awaits per-agent return series (Section 6). Finally, the demonstrated Sybil exploit needs a defense, not another demonstration: ranking must deduplicate or identity-weight the field before measured dispersion can be treated as adversary-resistant.

Coverage and observation. Single-name equities are absent for want of a keyless feed, gold because the public daily series were withdrawn, and two of nine datasets fail the realism battery and ship with the verdict recorded. The nine dataset-timeframe combinations reduce to roughly five independent market panels, all containing the 2020 and 2022 drawdowns, so the refusals of Section 5.3 are properties of the gates on this decade of data rather than of the gates alone. The agent sees twenty trailing closes; the protocol carries fields for fundamentals and news and the simulator leaves them empty, so this is a thinner information environment than FinBen or StockBench present.

The reliability gate is strict by construction. The default certifies profitability in every regime, and Sections 5.3 and 5.4 show that both it and a per-run drawdown bound refuse every long-only agent on every dataset containing a downturn. Whether either is the right definition of safe for a given mandate is the operator's choice, now exposed as configuration rather than hidden as a constant, and Section 3.2 states why the conservative default ships anyway; the paper does not claim to have found the right default. On short windows the per-run PSR bar is weak because the statistic scales with $\sqrt{n-1}$, which is a property of the PSR and not a choice. Eligibility, where it is ever granted, certifies a statistically survivable edge under stylized execution on narrow universes; it is not a deployability, capacity, or breadth certificate.

Honor-system inputs and field-level gaming. Declared trial counts are unverifiable and the measured dispersion is set by whoever populates the field; Section 3.3 states the threat model, the advisory status of declared-N, and the arena mitigations, none of which is claimed as built. Backtest verdicts for pretrained models are advisory by protocol rule because contamination, not multiple testing, is the binding threat there (Section 3.3); the shape-level contamination probe that would test this, scoring the same agent on real and moment-matched surrogate series, is specified but not run.

Built and not yet public. The arena now has one open operating window but no completed board: there is no hosted intake, scheduler or live board UI, so the league remains infrastructure an operator drives rather than a running service. External agents are sandboxed by container isolation only; multi-tenant hosting of untrusted submissions remains unbuilt. The perturbation diagnostic of Section 5.7 narrows the self-audit's blind spot but samples perturbations rather than searching adversarially, so a fragility that survives sampling can still hide. The calibration module's epistemic leg reads agreement among an agent's confidence streams as low uncertainty, so unanimous-but-wrong signals read as agreement and the leg is informative only when it reads high.

What changed during this paper. Three of the benchmark's own defects were found by writing it and are reported rather than hidden: a deflation prior in the wrong units (Section 5.2), a documentation claim that field-wide significance tests gated rank (Section 3.1), and an HMAC chain described as publicly verifiable when it was not (Section 4). This revision adds a fourth: the original luck floor degenerated into a bit-identical clone of buy-and-hold on one-symbol universes, so the falsification leg had no control on the rates dataset until the floor was redefined (Section 5.8). They are reported as findings and corrections because a benchmark paper that hides what its own evidence found is the thing the benchmark exists to prevent.

8 Conclusion

SharpeBench is an evaluation protocol that treats a trading agent's score the way a statistician would: deflated for search, required to survive every seed and window, zeroed on process violations, and tested against a serial-correlation-preserving null, with every number recomputable from committed data by a listed command. Applied to reference agents across nine frozen dataset-timeframe combinations, the gates refused everything and gave a legible reason each time; a controlled injected edge shows the bar sits near an annualized Sharpe of 3 on these window geometries, so the bar is strict rather than unsatisfiable. The thousand-agent floor leaves every zero-skill entrant ineligible, while the ninth audit case shows that field-measured deflation is not Sybil-resistant. Two of the paper's findings are corrections to the benchmark itself, and the general lesson of the first, that a deflation gate is only as correct as its unit conventions, applies to any scorer that mixes annualized literature constants with per-period statistics. The protocol and kernel are the contribution; the paid frontier-model field and the result of the already-open forward window are the next tests.

9 Reproducibility statement

The benchmark is a Rust workspace of twelve crates under the MIT and Apache-2.0 licenses, published to crates.io, npm and PyPI at the version stated in Appendix A, with a pinned toolchain and a lockfile. The scoring kernel forbids unsafe code. Every dataset in Table 1 is committed with a SHA-256 sidecar and a fetch script whose check mode reproduces the committed bytes from the public source. Every figure in this paper is produced by a committed script from the kernel's own formulas, and every table is produced by the commands in Appendix A from the committed data. Golden fixtures pin the full ranked output of two fields at full floating-point precision and are checked on Linux, macOS and Windows on every commit; a parity test asserts that WebAssembly and native scores are byte-identical. The self-audit runs on every commit: the build fails if any of eight claimed defenses regresses, or if the ninth case stops reproducing while it is still marked as an expected-vulnerable gap. Continuous integration for the commit this paper was built from is green on all twelve jobs.

No large language model is a component of the benchmark or of any result in this paper. The kernel has no judge. Language models were used to draft and edit prose; no citation in the bibliography was generated by one, and every arXiv entry was checked against its abstract page.

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A Commands that produce every number

All commands run from the repository root at the tagged version. The kernel is deterministic, so each command produces the same bytes on the three CI targets (Linux, macOS, Windows).

Version. Findings 1 and 2 were first observed at v0.2.1, and Table 3 was computed against v0.2.1 deliberately, since it documents that kernel's error. The units fix shipped in v0.3.0. Every other table and figure in this revision was regenerated on the v0.6.0 source tree; the original nine-dataset sweep remains byte-identical except for the documented single-symbol luck-floor correction, while the Sybil audit and thousand-agent file are new at v0.6.0.

The shipped demonstration (Figure 1a).

```
sharpebench score suites/example_submissions.json
```

The deflation curve (Figure 1b) and the demotion chart.

```
python paper/src/make-figures.py
```

The script reimplements the kernel's PSR, expected-maximum-Sharpe and normal-quantile functions in Python and states that it does; the figures are the kernel's formulas, not an independent computation.

The self-audit (Table 2).

```
sharpebench audit
```

The command reports eight defended cases and one expected-vulnerable Sybil case. A claimed defense that regresses or a known-gap marking that stops reproducing fails its commit-time tests.

Data realism (Table 1).

```
sharpebench realism --data data/<dataset>.csv
```

Regenerating any dataset and checking its hash.

```
python scripts/data/fetch_crypto.py --interval 1h --check
python scripts/data/fetch_fred.py fx --check
python scripts/data/derive_weekly.py --check
```

The evidence sweep (Table 4, Section 5.2, Section 5.3, Section 5.4, Section 5.8).

```
cargo run --release -p sharpebench-harness \
  --example evidence_sweep -- out.jsonl [dataset] [dsr_bar]
```

One JSON record per (dataset, configuration, agent), 512 per dataset. The grid, seeds, window rule, cost profile and both reliability configurations are constants in the source and are emitted on every record. The optional arguments restrict a run to one dataset or one deflation bar so the grid can be split across processes; because every seed is pinned, the concatenation of parts is identical to a serial run. The analysis that turns the records into the tables is a short script committed beside them under `paper/evidence/`.

The risk-managed agent (Section 5.5), its deflation N-sensitivity, and the perturbation report (Section 5.7).

```
cargo run --release -p sharpebench-harness \
  --example risk_managed_eval -- out.jsonl
```

Runs the risk-managed agent beside buy-and-hold and the luck floor on all nine datasets with the documented defaults, both reliability verdicts per row; emits the `n_sensitivity` records for weekly US indices at $N \in \{7, 10, 25, 50, 100\}$; and appends the perturbation spread report for weekly US indices. The fragile-agent perturbation separation quoted in Section 5.7 is produced by a unit test on synthetic data, not by this command:

```
cargo test -p sharpebench-harness perturb
```

The synthetic pass witness (Section 5.6).

```
cargo run --release -p sharpebench-harness \
  --example pass_witness -- pass-witness.jsonl
```

Sweeps a family of injected-edge agents (known per-period Sharpe by construction) beside a five-agent zero-edge field through the shipped default configuration on two window geometries, and writes one record per (shape, edge, agent).

The evidence figures (Figure 2a, Figure 2b).

```
python paper/src/make-evidence-figures.py
```

Every bar and point is a reduction over the committed records; no number is typed into the script.

The thousand-agent luck floor (Section 5.9).

```
cargo run --release -p sharpebench-harness \
  --example luck_floor_1000 -- \
  paper/evidence/final/luck-floor-1000.jsonl
```

The file contains one row per agent and one explicit summary row for each of the two daily datasets; the first five agents reproduce the shipped floor inside the same run.

The forward arena (Section 4).

```
sharpebench arena init <dir>
sharpebench arena open <dir> <window> <deadline> <reveal> [--config c.json]
sharpebench arena commit <dir> <window> <agent> <digest> <salt>
sharpebench arena advance <dir> <epoch>
sharpebench arena score <dir> <window> <data.csv> <reveals.json>
sharpebench arena publish <dir> <window> <key>
sharpebench arena verify <dir> [--pubkey <hex>]
```

Importing a rival field.

```
sharpebench import csv <dir-or-file> --out subs.json [--trials N]
sharpebench score subs.json
```

Golden fixtures and the parity test.

```
cargo test -p sharpebench-core --test golden_scores
cargo test -p sharpebench-sim  --test golden_input
cargo test -p sharpebench-wasm --test native_parity
```

Regeneration requires `SHARPEBENCH_UPDATE_GOLDEN=1` and refuses to run when `CI` is set.

Signing and verifying a board.

```
sharpebench sign   subs.json <hmac-key> board.json --ed25519 <secret>
sharpebench verify board.json --public
sharpebench verify board.json --pubkey <hex>
```

The second form verifies from the document alone. The third pins the key and rejects a board re-signed under a different one.

B Construct validity checklist

Bean et al. [2025] review 445 benchmarks and recommend that a benchmark paper answer an operational checklist. The answers below point to the section that supports each one, and say plainly where the answer is no.

Define the phenomenon. Yes. The phenomenon is skill that survives four corrections: deflation for the number of trials, reliability across seeds and windows, process discipline, and a bootstrap null. Section 2 states it and Equation (3) operationalizes it exactly.

Measure only the phenomenon. Partly. The eligibility predicate measures the phenomenon and nothing else. The scorer also reports fourteen statistics that do not gate, and Section 3.2 separates the two so a reader can tell which number moved the rank.

Construct representative datasets. Partly. Nine datasets across four asset classes and four bar sizes, each scored for realism before use (Appendix C.2). Single-name equities are absent, two datasets fail the realism battery and ship with the verdict recorded, and the observation is twenty trailing closes (Section 7).

Acknowledge limitations of reused datasets. Yes. Every dataset's source, license, caveats and failing stylized facts are in Table 1 and the data scripts. The commodities file keeps a negative oil close as published and documents the opt-out.

Prepare for contamination. Yes. Sealed held-out datasets under a published hash, a canary token, and a masked view that defeats ticker memorization (Appendix C.4). Forward commitment before the data exists (Section 4). The forward league that would exercise it has not run.

Use statistical methods. Yes. Every rank is gated on a deflated Sharpe with its number of trials, a block bootstrap with a fixed seed, and a per-run reliability test; field-wide Reality Check, SPA and step-down are reported; a bootstrap confidence interval on the DSR marks tied entries as tied. The annualization convention is stated once in Section 3.1 because getting it wrong was the first finding.

Conduct error analysis. Yes, and it is the paper. Section 5 traces every refusal to the gate that produced it and the quantity that justified it, and two of the four findings began as errors in the benchmark itself.

Justify construct validity. Partly, and the gap is named. The benchmark refuses every reference agent for reasons the data makes explicit, the luck floor never beats a reference agent, and the synthetic witness of Section 5.6 shows the acceptance region is nonempty and locates its boundary. What it has not shown is a real agent being admitted, so the claim that a real strategy can attain eligibility awaits the competing agent listed first in Section 7.

Statistical reporting. Every Sharpe in this paper is per period unless stated annualized, and Table 3 gives the conversion at every bar size. Every deflated Sharpe is reported beside its N. All runs use eight execution seeds and six windows; the per-cell record carries both. Costs are the typical profile of Appendix C. Block bootstrap, not i.i.d., throughout.

Licenses. The benchmark is MIT or Apache-2.0. FRED series are public domain. Binance klines are fetched from the public endpoint under its terms and redistributed as derived daily, 4-hour, hourly and weekly closes. No dataset requires a key.

Maintenance. The benchmark is versioned on crates.io, npm and PyPI with a pinned toolchain; a tag on the main branch publishes to every registry and a verification job confirms each serves the version. Datasets are frozen by hash and regenerated only by a deliberate script run. Golden fixtures change only by a local command that refuses to run in continuous integration. Adding a dataset means adding a hash-pinned file, a fetch script with a check mode, a realism verdict, and a periods-per-year value.

C Simulator, data, replay and contamination

C.1 Simulator and cost model

The data store returns rows at or before the decision index. Execution charges a per-trade fee, seeded slippage, own-order impact that scales with the square root of participation, and financing on gross exposure above one times net asset value. Fills can be capped at a fraction of participation with the remainder deferred. Three named cost profiles ship: none, typical, and worst case. The typical profile is 2 basis points fee, 3 slippage, 50 impact and 5 financing; worst case is 10, 15, 150 and 20 with a participation cap of 10 percent and a two-bar delay. All experiments in this paper use the typical profile. The seeded slippage is what makes different execution seeds produce different runs and gives pass^k something to measure.

C.2 Data

Nine frozen datasets ship with the benchmark, every one keyless and redistributable, each with a SHA-256 sidecar and a deterministic fetch script whose check mode reproduces the committed bytes. Table 1 in Section 3.4 lists them, including each dataset's frozen date range; the table is not repeated here. Before any agent is scored on a dataset, the dataset is scored: a battery of stylized facts after Cont [2001] tests excess kurtosis, volatility clustering, gain-loss asymmetry, aggregational Gaussianity and time-reversal asymmetry, and issues a verdict. Seven of the nine pass. Weekly US indices fail the aggregation test because 522 bars is too short for it, and FX majors fail time-reversal asymmetry, which is known to be weak in that market. Both ship with the verdict recorded. A realism gate that never fails is not a gate.

Two datasets carry caveats the scripts document. The commodities file keeps the WTI close of -36.98 on 20 April 2020 as published, which dominates its moments, so every commodities statistic in the paper inherits that outlier; a start date of May 2020 is the documented opt-out and removes it. The rates file's close column is a yield in percent, not a price. A Sharpe ratio on yield changes is therefore a statistic about a duration-unhedged directional rates position, not about any tradable instrument's return, and the dataset is best read as a stress case for the protocol on a one-symbol universe: cross-sectional momentum is undefined there, and the luck floor requires the single-symbol randomization of Section 5.8 to be a control at all. Single-name equities are absent because no keyless feed exists for them; that is the known gap in coverage.

C.3 Replay and recompute

A run can be captured as a trajectory holding only the agent's raw decisions, the window, and the execution seed, and never any return or self-reported statistic. A separate verifier replays the decisions through the same engine against the frozen dataset and recomputes the score. Capture and replay share one code path, so the round trip is exact by construction, and a test asserts that doubling every order in a captured trajectory recomputes to different returns. An artifact therefore cannot lie about its returns. What it cannot do is prove that the committed strategy produced those decisions; that is the job of the commitment in the next subsection, and the two mechanisms are not yet joined by code.

C.4 Contamination

Two defenses ship. A held-out dataset can be sealed under a published commitment to its hash, so the benchmark can prove later that the data a model was scored on is the data it committed to before scoring. A canary token with the standard training-corpus warning is embedded beside it, so post-hoc detection of a model that trained on the scenarios is possible. Separately, a masked view of any dataset renames symbols and dates to opaque identifiers, which defeats an agent that pattern-matches memorized tickers.

C.5 The units correction, in detail

The fix, shipped in v0.3.0, leaves the per-period kernel untouched and gives the configuration a periods-per-year field so that σ_{trials} is stated annualized and converted once at the point of use. A test asserts the conversion is applied exactly once on the configured path and never on the measured path. Another asserts that an SPX-like series is eligible under an annualized prior of 0.1 at 252 periods per year and ineligible under the same prior at one period per year, which reproduces the old units exactly. A test that the same series clears the original 0.5 prior was not written, because it cannot: 0.5 annualized at 50 trials is a benchmark of 1.14 annualized, and the index sits below it at any track length. That is the correct verdict for that prior, and the benchmark now states it in units a reader can check.